



SPDY speedier than HTTP

Google's SPDY – which has already been put to use in Chrome and Firefox 13 beta – is supposedly 23 per cent faster than HTTP <http://dgit.in/Isu6AR>

FB profile as identification?

Some nightclub security staff in the UK now demand access to clubbers' smartphones to verify identities <http://dgit.in/JidrnB>

From the Labs

stabilizer and the squiggle is the blur. The nature of the squiggle is the PSF and in this case it's a spatial PSF as it determines the path of the blur.

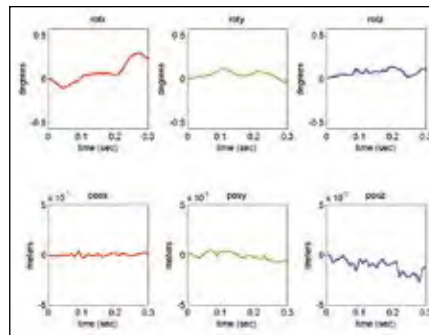
So how do you determine the PSF?

Advanced mathematics aside, the squiggle is determined by various algorithms. The algorithm tries to determine some reference points on the image captured (straight lines and corners are ideal), not to mention point sources of light and/or colour. Corners, such as the edges of doors and walls are especially useful as their motion can be very easily tracked and the PSF can be determined. In fact, when analyzing an image for the PSF, it's often advantageous to manually highlight an area where such pointers, including text, are present. This would narrow the search parameters and vastly reduce the required computing power. Other methods include image statistics, where a multitude of images are analysed and a pattern is inferred from them, which can then be replicated to apply to most average blur scenarios.

Deconvolution

The PSF is usually determined by a method known as blind deconvolution. What is convolution, you ask? Well, consider a pixel at a co-ordinate designated by (m,n). When this pixel is moved along a path (the PSF or blur path) determined by a function $f(x,y)$, it gets "convoluted" and merges with that function to create a new function $g(m,n)$. The purpose of deconvolution is to separate (m,n) and $f(x,y)$ from $g(m,n)$. The reason this is called "blind" deconvolution is that nothing is known other than the final output. The algorithm must determine $g(m,n)$; from that it has to find $f(x,y)$ and thus (m,n) and it has to do this for the millions of pixels that the image would probably consist of.

What agencies like NASA have done with its Hubble Space Telescope and similar projects is determine the PSF in advance. It can determine this PSF as it knows the exact shape of the lens and even the exact path of the telescope. Every movement is precisely documented during an exposure, similar to an inertial navigation system, and the PSF is generated. This PSF



Microsoft's sensors at work



The sensor used by Microsoft

is then simply fed to the algorithm and you end up with a clear image with a minimum of fuss and hassle.

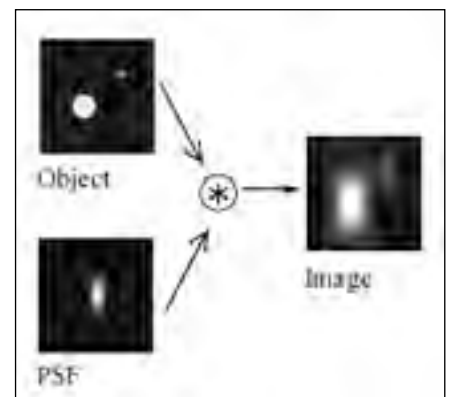
Microsoft's work

Microsoft Research has been hard at work since 2010, working on a hardware system to assist in this noble effort of deblurring your images. It developed a simple attachment consisting of sensors which analyse and track six specific parameters or degrees of freedom (DOF). These include factors such as acceleration, angular momentum, pitch and yaw, etc. These factors are meticulously recorded by the sensors (nothing more than standard accelerometers and gyroscopes) and a PSF is obtained. This PSF isn't 100 per cent accurate and is therefore only used to aid a modified blind deconvolution algorithm, creating what the researchers are calling an "aided-blind deconvolution" algorithm.

The science is still not perfect but it does go a long way in reducing and almost eliminating blur from an image. The hardware involved is very basic and given time and a degree of refinement, should be



Parameters measured by the sensor



Convolution (Source: Wikipedia)

compact enough to easily fit mobile phones and even camera lenses as a standard feature. The real advantage of this system, and where it beats camera stabilization, is in the fact that current IS only manages 2D stabilization – well short of the 6DOF system being developed by Microsoft.

A clearer future

The deblurring technology showcased by Adobe is still far from ready. Even the research being conducted by Microsoft is still in the prototype stage and there's a long way to go before this technology can get mature enough to see widespread usage. Having said that, what we've covered in this article barely scratches the surface of what has and is being done in this field. That future however is something we're looking forward to with a lot of anticipation. Taking photographs will just be so much easier, just imagine, fancy lenses and tripods can easily become relics of the past. Photography can return to its roots, where all the photographer has to worry about is the scene in question rather than the gear to be carried. 